Check: vabs(-4) returns 4.

4. Back · Flash [NUM] Write in decimal notation 2.5×10^3 then 6.2×10^{-2}.

Ans. 2 500 ; 0,062.

5. Running thread · Flash [REP] Give the image of $A(2;-3)$ under point reflection in the origin.

Ans. $(-2;3)$.

Sheet 79 Thu. 10 June 2027 Presentation of the specialist subjects; review

1. Warm-up · Flash [NUM] Express $\frac{3}{8}$ as a percentage.

Ans. 37,5 %.

2. Core · Metacognition [MET] What are your three revision priorities before Première?

Ans. Individual answer: find in the year's sheets the questions where you hesitated most.

3. Core + · Flash [EQ] Solve $x^2 = 18$.

Ans. $x^2 = 9 : x = 3$ or $x = -3$.

4. Back · Trap [PUI] Calculate $(-2)^4$ and -2^4.

Ans. 16 and -16.

The brackets decide what the exponent applies to.

5. Running thread · True/false [DIV] True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.

Ans. False. Counterexample: 12 is divisible by 4 and by 6, not by 24; it is divisible by LCM $(4\,;\,6) = 12$.

Annual review and consolidation of Seconde learning

Sheet 80 Mon. 14 June 2027 Annual review and consolidation

1. Warm-up · Flash [GEO] $A(-1;4)$ and $B(3;1)$. Coordinates of $\overrightarrow{AB}$?

Ans. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

2. Core · Method [GEO] Slope-intercept equation of (AB) with $A(0;1)$ and $B(2;5)$.

Ans. Gradient 2; $y = 2x + 1$.

3. Core + · Method [REP] $A(1;0)$, $B(4;1)$, $C(5;4)$. Determine D so that $ABCD$ is a parallelogram.

Ans. $D = A + C - B = (2;3)$; check: $\overrightarrow{AB} = \overrightarrow{DC} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

4. Back · Flash [CAL] Calculate $-5 + 8 \times (-2)$.

Ans. $-5 - 16 = -21$.

Multiplication takes priority.

5. Running thread · Statement→ formula [REG] The area of a square with side x is a function A of x. Give $A(x)$, a table of values for $x = 1; 2; 3$, and say whether A is proportional to x.

Ans. $A(x) = x^2$; 1; 4; 9; not proportional: $A(2) \neq 2A(1)$.

Sheet 81 Tue. 15 June 2027 Annual review and consolidation

1. Warm-up · Flash [NUM] Write 0.00045 in standard form.

Ans. $4{,}5 \times 10^{-4}$.

2. Core · Flash [NUM] Determine $[-2;3[\cap]0;5]$.

Ans. $]0;3[$.

3. Core + · Method [EQ] Solve $|x - 1| < 4$ and give S as an interval.

Ans. $-4 < x - 1 < 4 : S =]-3;5[$.

4. Back · Expand [DEV] Expand and simplify $(2x - 1)^2$.

Ans. $4x^2 - 4x + 1$.

5. Running thread · Flash [GEO] A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.

Ans. $V = \pi \times 3^2 \times 5 = 45\pi$ cm^3.

Sheet 82 Thu. 17 June 2027 Annual review and consolidation

1. Warm-up · Flash [IMG] $f(x) = x^2 - 4x$. Calculate $f(3)$.

Ans. $9 - 12 = -3$.

2. Core · Flash [EQ] Solve $x^2 = 9$ then $|x| = 2$.

Ans. $x = \pm 3$; $x = \pm 2$.

3. Core + · Method [REG] $f(x) = 2x - 3$ and $g(x) = x + 1$. Solve $f(x) < g(x)$.

Ans. $2x - 3 < x + 1 \Leftrightarrow x < 4 : S =]-\infty; 4[$.

4. Back · Flash [QP] A tap delivers 18 L in 4 min. What volume in 10 min?

Ans. $4{,}5 \times 10 = 45$ L.

5. Running thread · Reading code [ALG] What is the final value of `n` ?

```
n = 0
while n*n < 30:
    n = n + 1
```

Ans. $n = 6$ since $5^2 = 25 < 30$ but $6^2 = 36 \geqslant 30$.

Smallest integer whose square exceeds 30.

Sheet 83 Mon. 21 June 2027 Annual review and consolidation

1. Warm-up · Flash [STA] Mean of 5, 7, 9, 11?

Ans. 8.

2. Core · Method [STA] Data set 1; 3; 4; 6; 8; 9; 11: Q_1, median, Q_3.

Ans. $n = 7$: Q_1 index $\lceil 7/4 \rceil = 2$: 3; median (index 4): 6; Q_3 index $\lceil 21/4 \rceil = 6$: 9.

3. Core + · True/false [STA] True or false? Justify. Adding 10 to each value of a data set leaves its standard deviation unchanged.

Ans. True. The deviations from the mean do not change.

4. Back · Reflection [IMG] How many preimages do -1 and 4 have for $f(x) = x^2$?

Ans. -1: none; 4: two (-2 and 2).

5. Running thread · Prove [LIT] Show that for every real number x : $(x-1)(x+1) + 1 = x^2$.

Ans. $(x-1)(x+1) + 1 = x^2 - 1 + 1 = x^2$.

Sheet 84 Tue. 22 June 2027 Annual review and consolidation

1. Warm-up · Flash [DEV] Expand and simplify $(2x - 3)(x + 4)$.

Ans. $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.

2. Core · Flash [EQ] Solve 3 $x - 7 \leqslant 2x + 5$.

Ans. $x \leqslant 12$; $S =]-\infty; 12]$.

3. Core + · Method [EQ] Solve $(x-1)(x+4) = 0$ then $x^2 = 5$.

Ans. $x = 1$ or $x = -4$; $x = \sqrt{5}$ or $x = -\sqrt{5}$.

4. Back · Reflection [PNP] Subscription: 5 € + 3 € per entry. Is the price proportional to the number of entries?

Ans. No: 1 entry $\mapsto$ 8 €, 2 entries $\mapsto$ 11 $\neq$ 16 €.

5. Running thread · Flash [SOL] A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.

Ans. Slant height $\sqrt{3^2 + 4^2} = 5$ cm; $V = \frac{1}{3}\pi \times 9 \times 4 = 12\pi$ cm^3.

Sheet 85 Thu. 24 June 2027 Annual review and consolidation

1. Warm-up · Flash [PRO] What multiplier corresponds to an 8% increase?

Ans. 1,08.

2. Core · Method [PRO] A 25% increase is followed by a 20% decrease. What is the overall rate?

Ans. $1{,}25 \times 0{,}8 = 1$: no change (0%).

3. Core + · Flash [PRO] A price goes from 5,000 to 5,750 Ar. Rate of change?

Ans. $\dfrac{750}{5\,000} = 15\,\%$.

4. Back · Recognise [PER] What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?

Ans. A rectangle; a disc.

5. Running thread · Problem [EQ] Three times a number, minus 4, equals twice that number plus 5. Write an equation then find the number.

Ans. $3x - 4 = 2x + 5$, hence $x = 9$.
Check: $3 \times 9 - 4 = 23 = 2 \times 9 + 5$.

Sheet 86 Mon. 28 June 2027 Annual review and consolidation

1. Warm-up · Flash [MOD] Direction of variation of $x \mapsto -3x + 2$?

Ans. Strictly decreasing (gradient $-3 < 0$).

2. Core · Method [REG] Minimum and maximum of $f(x) = x^2$ on $[-3;1]$?

Ans. Minimum 0 (at 0); maximum 9 (at -3) ; $f(1) = 1 < 9$.

3. Core + · Method [CMP] For $x > 0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.

Ans. $0 < x < x + 1$ and the reciprocal function is decreasing on $]0;+\infty[$: $\dfrac{1}{x} > \dfrac{1}{x+1}$.

4. Back · Construction steps [CON] Describe in three steps how to construct the reflection A' of A by relative to a line (d).

Ans. 1. Perpendicular to (d) through A, cutting (d) in H. 2. Transfer with compasses HA on the other side of (d). 3. A' is the point obtained: H is the midpoint of $[AA']$.

5. Running thread · Method [DEM] A point I satisfies $IA = IB$. Where is it? Which property?

Ans. On the perpendicular bisector of $[AB]$ (perpendicular bisector property).

Sheet 87 Tue. 29 June 2027 Annual review and consolidation

1. Warm-up · Flash [PRB] $P(A) = 0.28$. Calculate $P(\overline{A})$.

Ans. 0,72.

2. Core · Method [PRB] 100 people including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.

Ans. $P_F(G) = \dfrac{6}{60} = 0{,}1$; $P_G(F) = \dfrac{6}{10} = 0{,}6$.

3. Core + · Method [PRB] $P(A) = 0{,}4$, $P_A(B) = 0{,}5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.

Ans. $0{,}4 \times 0{,}5 + 0{,}6 \times 0{,}2 = 0{,}2 + 0{,}12 = 0{,}32$.

4. Back · Problem [MOD] Volume of a cube with edge x : $V(x) = x^3$. Calculate $V(3)$ and find x such that $V(x) = 64$.

Ans. $V(3) = 27$; $x^3 = 64$, $x = 4$.

5. Running thread · Flash [REP] In which quadrant (numbered I to IV anticlockwise) is the point $M(a;b)$ with $a < 0$ and $b > 0$?

Ans. Quadrant II (top left).

Sheet 88 Thu. 1 st July 2027 Annual review and consolidation

1. Warm-up · Flash [GEO] Hypotenuse of a right-angled triangle with shorter sides 5 and 12?

Ans. $\sqrt{25+144}=13$.

2. Core · Flash [REP] Midpoint of $[AB]$ with $A(-3;2)$ and $B(5;-4)$?

Ans. $\left(\frac{-3+5}{2};\frac{2-4}{2}\right)=(1;-1)$.

3. Core + · Write up [DEM] Show that $A(0;0)$, $B(2;1)$, $C(-1;2)$ form a triangle that is right-angled and isosceles at A.

Ans. $AB^2=5$, $AC^2=5$; $BC^2=9+1=10=AB^2+AC^2$: converse of Pythagoras' theorem.

4. Back · Flash [NUM] Give an approximate value of $\frac{1}{3}$ à 10^{-2} accuracy. Is this number a decimal?

Ans. 0.33. No: the decimal expansion of $\frac{1}{3}$ is non-terminating (0.333 . . .).

5. Running thread · Problem [DIV] What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, greater than 1?

Ans. $1+\text{LCM}\ (4\,;6)=13$.